

Introduction to R programming language

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Advanced Data Analysis (ADA) Unit

12th January 2026

Course Outline

- **Day 1**

- R Basics – Operations, Packages & Classes
- Loops & Conditions
- Data Manipulation & Plotting
- Functions in R – From use to development

- **Day 2**

- Best Practices (Documentation, Reproducibility, ...)
- AI & Coding
- tidyverse Universe



Programming & Research

- **Transparency**
- **Reproducibility**
- **Up-to-date methodology**
- **Big Data Era**



Programming & Researcher

- **Documentation**
 - Citations!
 - Job market!
- **Capability & Flexibility**
- **Understanding Data Analysis**

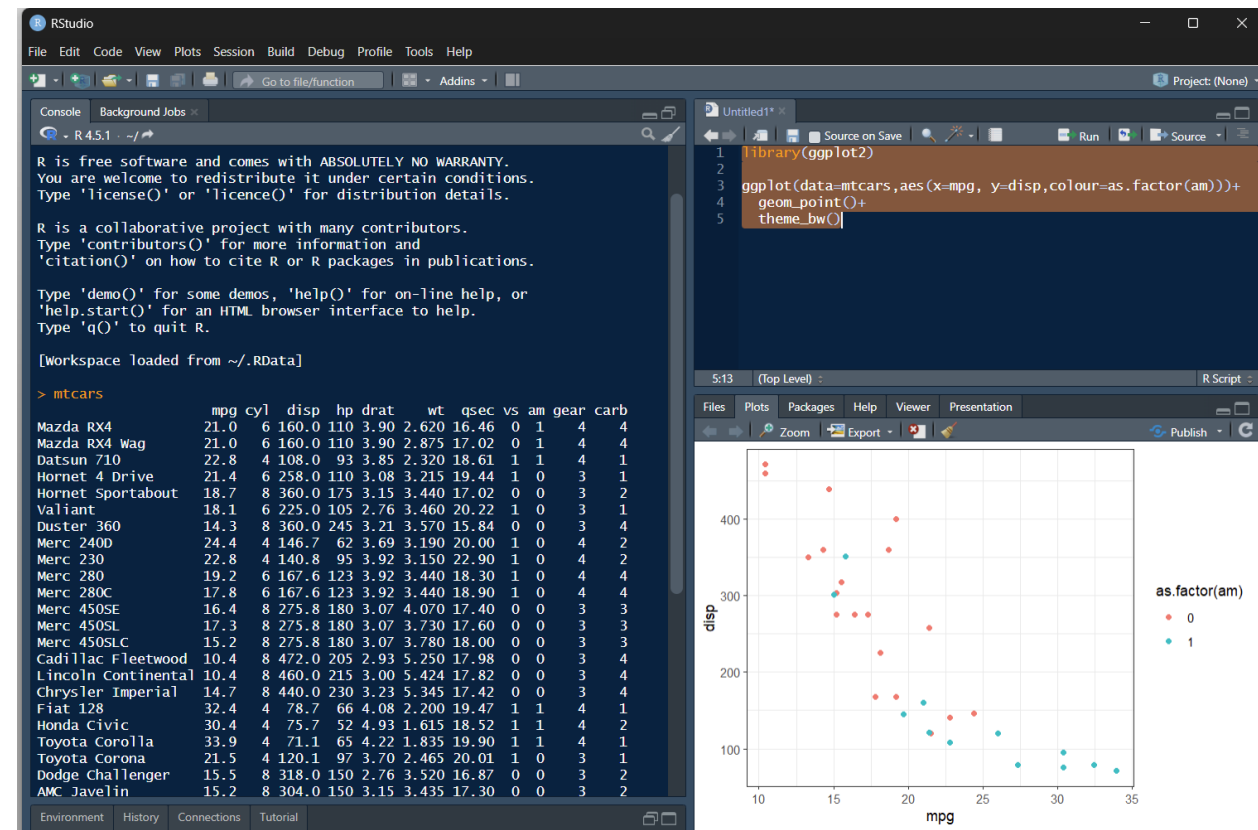


Why R



- Statistical Analysis & Data Visualization
- Free-to-use, Open Source
- Thousands of curated & up-to-date packages

Integrated Development Environment (IDE)



Why R



- Statistical Analysis & Data Visualization
- Free-to-use, Open Source
- Thousands of curated & up-to-date packages

Literate Programming



Preliminary Assessment

We are going to use the DESeq2 package, introduced in [@love2014](#).

Here, one can make a plot, for example, of log2 intensities

```
{r,fig.width=8,fig.height=6,fig.dpi=300}
# label: FCImport
# code-summary: File Import & Initial Assessment
# results: asis

counts = read.table("data/03_feature_counts/results_custom_full/special_results
.txt",
                    header=T,row.names = 1) %>%
  select(-c("Chr","Start","End","Strand","Length"))

colnames(counts) = c("vdd1","vdd2","vdd3","vdd4",
                    "wt1","wt2","wt3","wt4")

threshold = 5

zero_counts = rowMeans(counts)>0
cutoff_counts = rowMeans(counts)>threshold

cat(
  paste0("***Total Genes: **", nrow(counts), " genes\n\n",
        "***Genes with non-zero counts:** ", sum(zero_counts), " genes (",
        round(100*sum(zero_counts)/nrow(counts),2), "%)\n\n",
        "***genes above threshold** (mean counts >", threshold,")**:",
        sum(cutoff_counts), " genes (",
        round(100*sum(cutoff_counts)/nrow(counts),2), "%)\n\n")
)
```

Preliminary Assessment

We are going to use the DESeq2 package, introduced in Love, Huber, and Anders ([2014](#)).

Here, one can make a plot, for example, of log2 intensities

▼ File Import & Initial Assessment

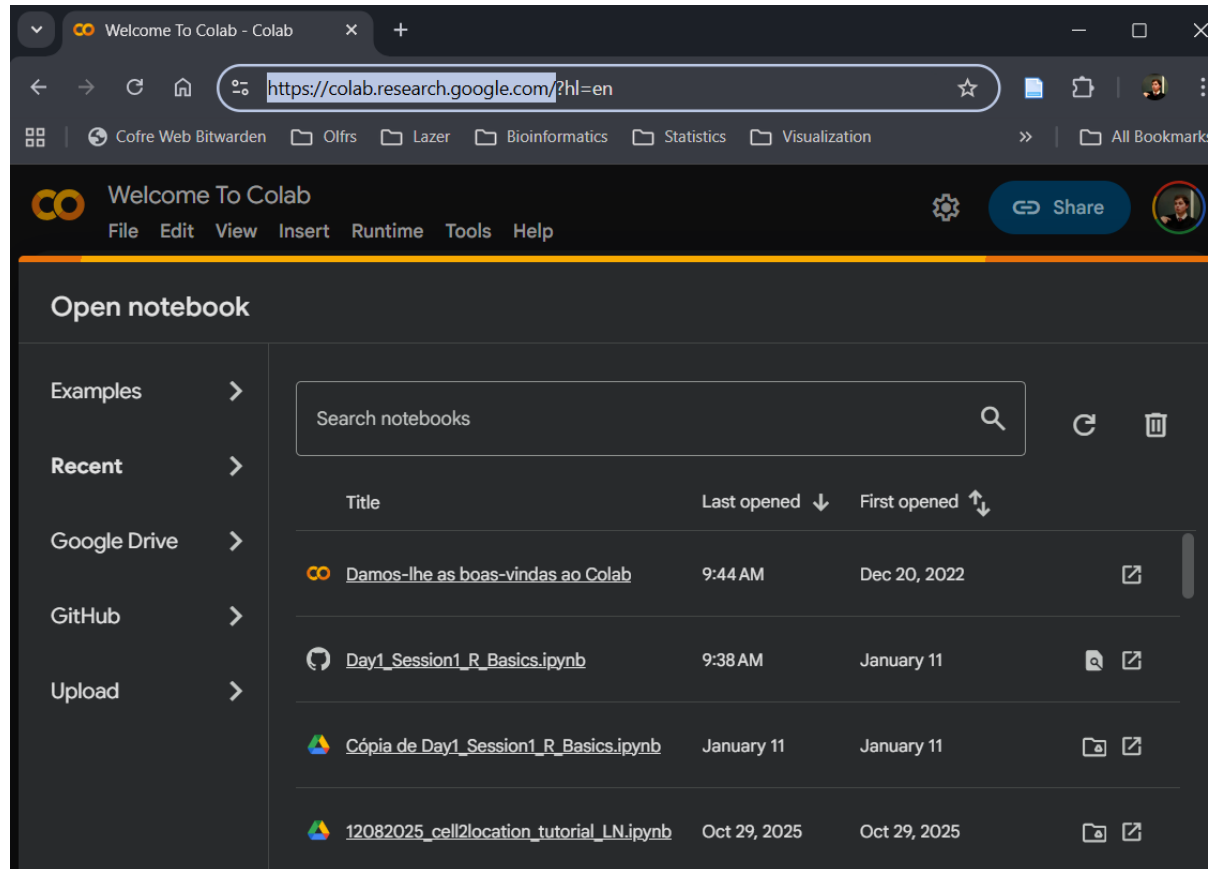
```
1 counts = read.table("data/03_feature_counts/results_custom_full/special_results.txt",
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14         "***Genes with non-zero counts:** ", sum(zero_counts), " genes (",
15         round(100*sum(zero_counts)/nrow(counts),2), "%)\n\n",
16         "***Genes above threshold** (mean counts >", threshold,")**:",
17         sum(cutoff_counts), " genes (",
18         round(100*sum(cutoff_counts)/nrow(counts),2), "%)\n\n")
19 )
```



Course Material

https://github.com/andreboilerbarros/Intro_R

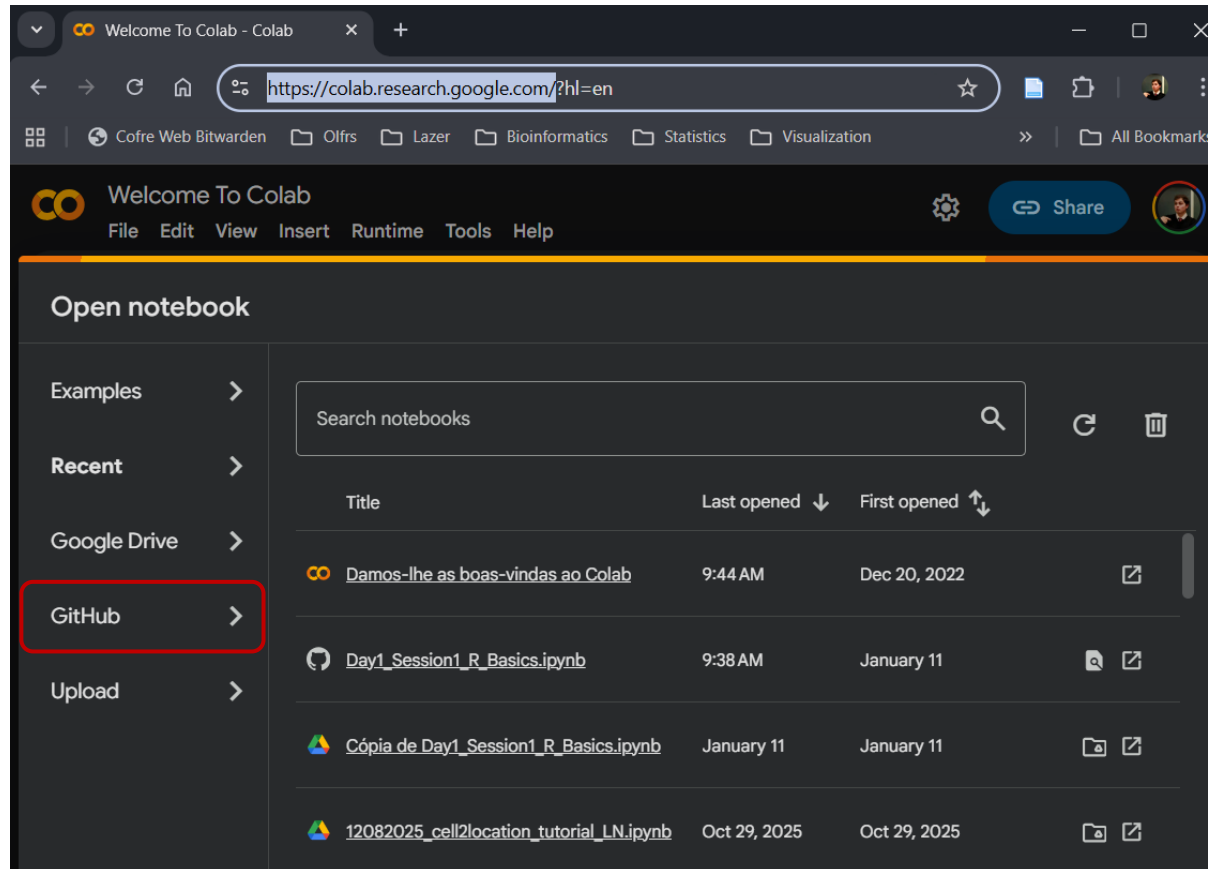
1. Google Colab – colab.research.google.com



Course Material

https://github.com/andrebolbarros/Intro_R

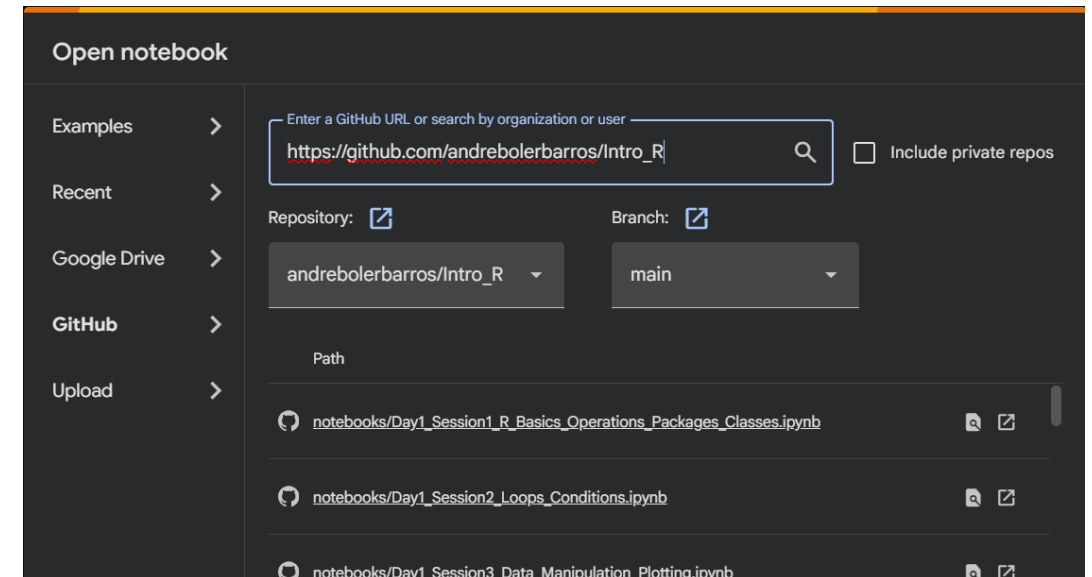
1. Google Colab – colab.research.google.com



Open notebook

Search notebooks

Title	Last opened ↓	First opened ↑	
Damos-lhe as boas-vindas ao Colab	9:44 AM	Dec 20, 2022	
Day1_Session1_R_Basics.ipynb	9:38 AM	January 11	
Cópia de Day1_Session1_R_Basics.ipynb	January 11	January 11	
12082025_cell2location_tutorial_LN.ipynb	Oct 29, 2025	Oct 29, 2025	



Open notebook

Enter a GitHub URL or search by organization or user

https://github.com/andrebolbarros/Intro_R

☐ Include private repos

Repository: Branch:

andrebolbarros/Intro_R main

Path

notebooks/Day1_Session1_R_Basics_Operations_Packages_Classes.ipynb

notebooks/Day1_Session2_Loops_Conditions.ipynb

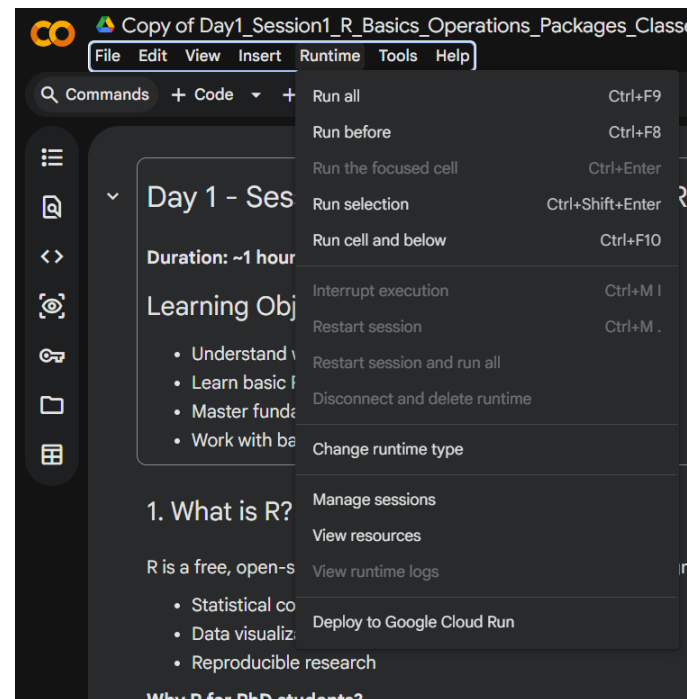
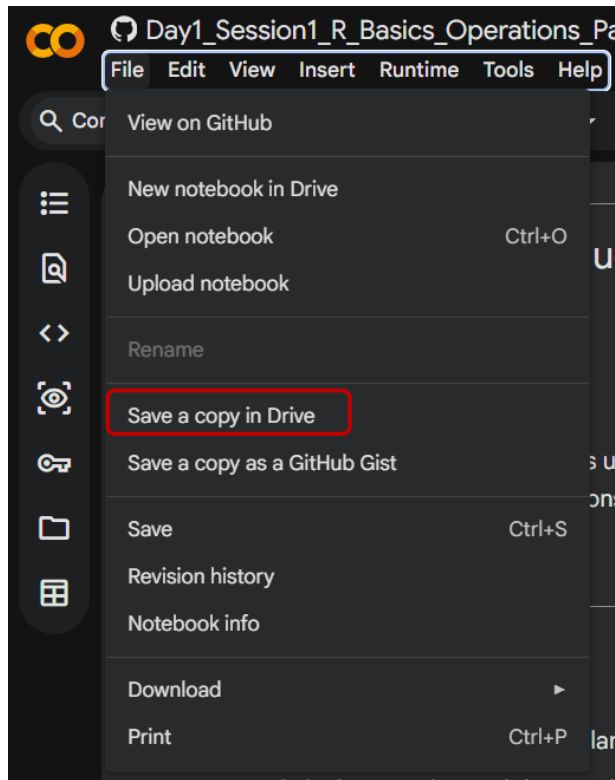
notebooks/Day1_Session3_Data_Manipulation_Plotting.ipynb



Course Material

https://github.com/andrebolbarros/Intro_R

2. Save your own copy



https://github.com/andrebolbarros/Intro_R

3. Change kernel (language)

